

CHAPTER 1

Oceans and Life

NCERT Exercise Solutions

The Big Questions	5 Q
Let's Explore · Ocean Relief	3 Q
Let's Explore · Waves	1 Q
Think About It · Ocean Currents	1 Q
Let's Explore · Ocean Currents	1 Q
Let's Explore · Cyclones and Tsunamis—Early Warning Systems	1 Q
Think About It · Early Warning	1 Q
Let's Explore · International Maritime Rules and Regulations	1 Q
Questions and activities	9 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

? The Big Questions

1 What is the importance of ocean relief?

✔ ANSWER

- Ocean relief means landforms on the ocean floor: continental shelf, slope, seamounts, abyssal plains, mid-ocean ridges, trenches.
- It influences marine life, ocean movements and where resources are found (the continental shelf is rich in marine life, oil and gas).
- It matters for fishing, shipping routes and undersea exploration.

2 How does ocean water movement impact life and environment?

✔ ANSWER

- Waves shape coastlines by erosion (caves, arches, stacks) and deposition (beaches).
- Tides help ships enter harbours, support fishing and can make electricity.
- Currents carry warm or cold water and affect the climate of coasts (e.g. the Gulf Stream keeps Norway's waters warm).
- Where warm and cold currents meet, rich fishing grounds form.

3 **Why are maritime zones important for a country?**

✔ ANSWER

- They show how far a country's rights at sea extend: territorial sea (12 NM), contiguous zone (24 NM), EEZ (200 NM).
- The country can make laws on customs, immigration and pollution in its territorial waters.
- In its EEZ it has special rights to use fish, oil and gas.
- Clear zones under UNCLOS reduce disputes between countries.

4 **How can we use ocean resources in a sustainable way?**

✔ ANSWER

- Follow the Blue Economy idea: use ocean resources for growth and jobs while keeping marine ecosystems healthy.
- Reduce marine pollution, regulate fishing, conserve marine areas (SDG 14: Life Below Water).
- Follow international rules such as MARPOL and cooperate with other countries.

5 **How do early warning systems help protect people from tsunamis and cyclones?**

✔ ANSWER

- Seismic stations, ocean bottom pressure recorders, tide gauges and satellites detect and track hazards.
- Alerts reach governments and people quickly through satellite, SMS, emails and coastal display boards.
- People get time to evacuate, so fewer human and animal lives are lost.
- Examples: ITEWC tsunami buoys, SSEWS storm-surge warnings, GEMINI device for fishermen.

⭐ Let's Explore · Ocean Relief

1 **Looking at Fig. 1.2: How do you think these oceanic features would have been formed?**

💡 SAMPLE ANSWER

- Mid-ocean ridges, seamounts and volcanic islands are formed by volcanic activity (magma rising from below).
- Trenches and rift valleys form where the Earth's crust sinks or pulls apart.
- Guyots are seamounts flattened by wave action over a very long time.

- Abyssal plains are flat because sediments settle and cover the deep floor.

2 **Which part of the ocean floor appears deep and shallow?**

✔ ANSWER

- **Shallow:** the continental shelf, near the coast.
- **Deep:** the abyssal plain and, deepest of all, the ocean trenches (e.g. Mariana Trench).

3 **How do you think these underwater features would have an effect on marine life and ocean movements?**

💡 SAMPLE ANSWER

- The shallow, sunlit continental shelf is rich in marine life and fish.
- Ridges, seamounts and slopes change the direction and speed of currents and waves.
- Deep trenches are dark and cold, so only special life forms live there.

⭐ Let's Explore · Waves

1 **Find out the names of some famous sea beaches in our country and locate them on the map of India.**

💡 SAMPLE ANSWER

- Marina Beach – Chennai, Tamil Nadu
- Juhu Beach – Mumbai, Maharashtra
- Calangute and Baga Beaches – Goa
- Kovalam Beach – Keralam
- Puri Beach – Odisha
- Radhanagar Beach – Andaman & Nicobar Islands

(Mark these places on an outline map of India.)

🤔 Think About It · Ocean Currents

1 **Observe the map carefully and note the movement patterns of cold and warm currents. Do you think these currents affect the temperature of nearby coastal regions?**

✔ ANSWER

Yes.

- Warm currents flow from warm regions and make nearby coasts warmer (Gulf Stream keeps Norway's waters warm; sea ice does not form easily there).

- Cold currents make coasts colder (Labrador Current helps sea ice and icebergs form off Canada even at lower latitudes).

⭐ Let's Explore · Ocean Currents

1 **Observe Fig. 1.9 and take a note on ocean currents along the coastline of India. What is the nature of ocean currents?**

💡 SAMPLE ANSWER

- India lies in the northern Indian Ocean, close to the Equator, where the map shows the warm Equatorial Counter Current and currents circulating in the Arabian Sea and Bay of Bengal.
- The currents near India are mostly **warm** currents.
- They change direction with the monsoon winds (seasonal currents) (check the map in your book).

⭐ Let's Explore · Cyclones and Tsunamis—Early Warning Systems

1 **Simeulue Island was largely spared in the 2004 tsunami partly because of local folklore. Talk to your elders about similar folk traditions or stories on natural disasters in your culture and discuss how traditional knowledge can help communities stay safe.**

💡 SAMPLE ANSWER

- Example: elders in coastal areas say that if the sea suddenly goes far back from the shore, people must run to high ground at once.
- Other signs told in stories: animals and birds behaving strangely, ants moving to higher places before heavy rain.
- Such knowledge, passed down through generations, helps people recognise danger early and act quickly, even without modern alerts.

🤔 Think About It · Early Warning

1 **If you receive a tsunami or cyclone alert on your phone right now, do you know the five steps your family should take to stay safe?**

💡 SAMPLE ANSWER

- Stay calm and follow official alerts (radio, TV, SMS).

- Keep an emergency kit ready: water, food, medicines, torch, documents, charged phone.
- Move to high ground or the nearest cyclone shelter as directed; leave low-lying coastal areas.
- Switch off electricity and gas; stay indoors away from windows during a cyclone.
- Do not return until officials say it is safe.

⭐ Let's Explore · International Maritime Rules and Regulations

1 **Can you find out some countries that have conflicts over their territorial sea waters?**

💡 SAMPLE ANSWER

- China and the Philippines, Vietnam, Malaysia – South China Sea
- China and Japan – East China Sea
- Greece and Türkiye – Aegean Sea
- Israel and Lebanon – Mediterranean Sea (sea boundary)

❓ Questions and activities

1 **How do different features of the ocean floor affect human activities such as fishing, transport, and underwater exploration?**

✔ ANSWER

- **Fishing:** the shallow continental shelf is rich in marine life, so most fishing happens there.
- **Resources:** the shelf also has oil and gas.
- **Transport:** ships need enough depth; shallow areas (e.g. Singapore Strait) force large ships to wait for high tide.
- **Exploration:** trenches, ridges and seamounts are studied through bathymetry; deep areas are hard to explore.

2 **How do waves, tides, and ocean currents influence climate, marine life, and human activities around the world?**

✔ ANSWER

- **Waves** shape coastlines (caves, arches, beaches); storm waves and tsunamis damage coasts.
- **Tides** help ships enter harbours, bring fish close to shore during high tide, and can generate electricity (Gulf of Kutch, Gulf of Khambhat, Sundarbans).
- **Currents** affect climate: the warm Gulf Stream keeps Norway's waters warm; the cold Labrador Current brings sea ice.
- Where warm and cold currents meet, waters are rich in nutrients and form fishing grounds.

3 **Describe the importance of marine resources.**

✔ ANSWER

- **Food and livelihood:** fish give protein; fishing and aquaculture give jobs.
- **Energy:** offshore oil and gas; 'blue energy' from tides and waves.
- **Minerals:** sand, gravel and deep-sea minerals.
- **Trade and tourism:** 90% of global trade by volume goes by sea; coastal tourism is a big industry.
- **Environment and culture:** support biodiversity; śhaṅkha and cowries link the sea with prosperity and culture.

4 **Draw a poster that encourages people to protect marine resources. Include a slogan (like Save Our Seas or Fish for the Future) and explain your message in 2–3 sentences.**

💡 SAMPLE ANSWER

Slogan: **"Healthy Oceans, Healthy Future!"**

Picture idea: a clean blue sea with fish and coral on one side and plastic waste on the other.

Message: Oceans give us food, jobs and clean air. Plastic and over-fishing are killing marine life. Do not throw waste in the sea and catch only what we need.

5 **Using a map of India, mark at least five coastal states that might need tsunami warnings.**

💡 SAMPLE ANSWER

- Tamil Nadu
- Andhra Pradesh
- Keralam
- Odisha
- West Bengal
- also Puducherry and the Andaman & Nicobar Islands (UTs badly hit in 2004)

(Mark them on an outline map of India.)

6 **Work in a group and act out what would happen when a cyclone warning is issued. Highlight what role different people, like scientists, officials, and coastal villagers, would play.**

💡 SAMPLE ANSWER

- **Scientists (IMD/INCOIS):** track the cyclone by satellite, forecast its path and storm surge, issue the warning.

- **Officials:** pass the alert by SMS, radio and loudspeakers, open cyclone shelters, arrange buses, rescue teams, food and medicines.
- **Fishermen:** get alerts on GEMINI, return to shore, secure boats.
- **Villagers:** pack an emergency kit, move with children, elders and animals to shelters, help neighbours.

7 **Suppose a foreign ship is sailing through India's territorial sea. What principle allows the ship to pass through these waters, subject to certain conditions?**

✔ ANSWER

Innocent Passage.

- The ship may pass if it does not threaten India's peace, good order or security.
- It must move continuously without stopping (except in emergencies or breakdowns).
- It cannot fish, carry out military operations or other non-peaceful actions.

8 **A country has an Exclusive Economic Zone (EEZ) extending up to 200 nautical miles from its baseline. What rights does the country have over the natural resources in this zone?**

✔ ANSWER

- Under UNCLOS, the country has **special rights to use the ocean resources** of its EEZ, such as fish, oil and gas.
- But it does not fully own or control the waters: ships of other countries may still pass peacefully.

9 **Create a model of an ocean with ship routes, protected zones, and safety measures. Based on the model, explain how it follows maritime laws and protects the ocean.**

💡 **SAMPLE ANSWER**

Model idea: a tray with blue clay/paper showing a coast, baseline, 12 NM territorial sea, 24 NM contiguous zone, 200 NM EEZ and the high seas.

- **Ship routes** marked with lanes and lighthouses: ships follow COLREGs (lookout, safe speed, lights and signals) to avoid collisions.
- **Protected zones** around coral reefs and fishing grounds: no fishing or drilling, as SDG 14 asks.
- **Safety measures:** tsunami buoys, tide gauges and warning centre; ships do not dump waste (MARPOL).
- Foreign ships pass the territorial sea under innocent passage; the country uses resources in its EEZ.

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**